Four teams. Four revenue numbers.
Zero executive trust.

30
daily CSV files ingested
30 files · 4 regions · 12 categories

4
DQ failure categories

$9.8M
revenue under management

0
agreed revenue number

Revenue was calculated differently by Finance, Ops, Product, and Leadership — from identical raw files.

APPROACH

A modular, lightweight data platform

WHAT WE BUILT

Raw CSV Files
30 daily sales files · multi-region

Python Ingestion
Idempotent · duplicate prevention

DQ Validation Layer
Schema · nulls · freshness · dupes

DuckDB Warehouse
Lightweight · local-first · fast

SQL / DBT Models
Revenue aggregations · transformations

Streamlit Dashboard
Executive KPIs · filters · exports

Python | DuckDB | SQL | Streamlit | Plotly

KEY TRADE-OFFS

Why DuckDB over Postgres or BigQuery?
Zero infra overhead. Columnar speed for analytics. Production-ready for this data volume.

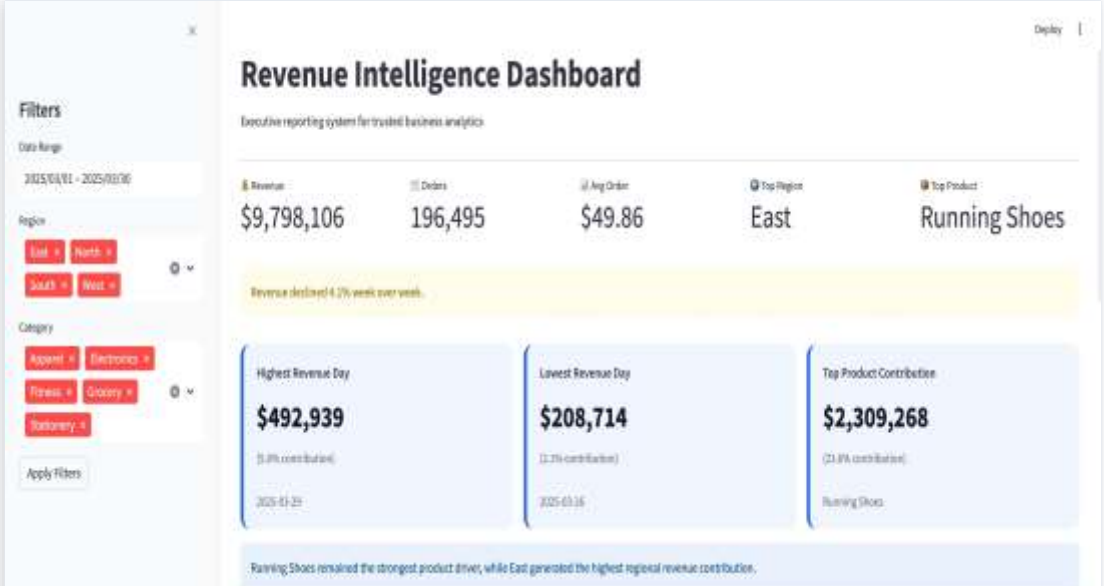
Idempotent reruns — how?
Each load checks order_id presence before insert. Re-running 30 files yields identical results.

Why modular DQ checks?
Each check is isolated — schema, nulls, freshness, duplicates. Failures don't cascade.

Local-first or cloud-ready?
Local-first now. Architecture is designed for Airflow + Snowflake migration in next phase.

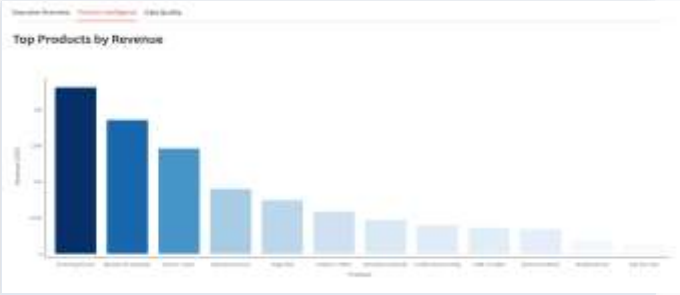
THE BUILD

Live dashboard built on validated, trusted data



Executive KPI Overview — Revenue · Orders · Avg Order · Regional Split

Revenue Trend + 7-Day Moving Average



Data Quality Monitoring — DQ Risk Score 90/100

- ✓ Moving Avg Trend
- ✓ Duplicate Detection
- ✓ Regional Breakdown
- ✓ Freshness Checks

OUTCOMES

Measurable results, engineered in.

30

Daily files ingested

Idempotent, every run

4

DQ issue categories caught

Schema · nulls · freshness · dupes

0

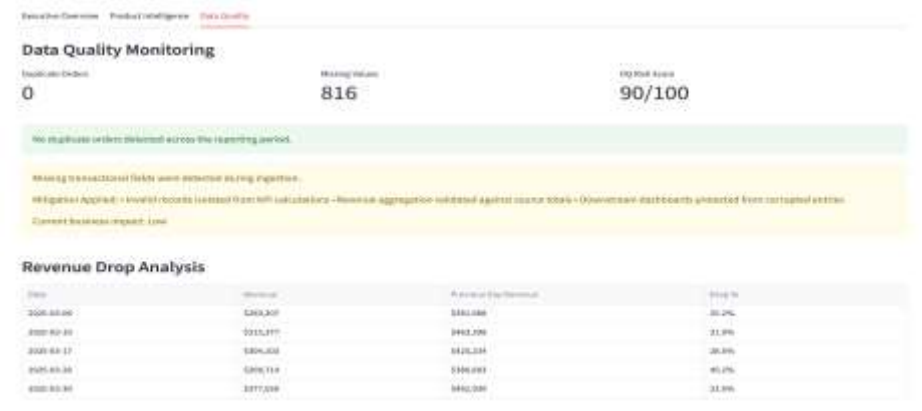
Duplicate revenue inflation

After idempotent reruns

90/100

DQ Risk Score

Validated & monitored



- Unified revenue logic — one calculation, everywhere
- Trusted executive reporting layer
- Data quality visibility for the first time
- Pipeline runs clean from day 1

WHAT'S NEXT

With another week, here's what gets built.

Day 1–2

Airflow Orchestration

Schedule daily CSV ingestion. Alert on pipeline failure. Retry logic with backoff.

Highest priority — production-grade reliability

Day 3–4

Cloud Warehouse Migration

Move DuckDB to Snowflake or BigQuery. Unlock multi-user access and row-level lineage.

Architecture already designed for this

Day 4–5

Automated Alerting

Slack or email alerts on DQ failures, revenue anomalies, and stale data ingestion.

Connects monitoring to action

Day 5–7

Revenue Forecasting

Rolling average baseline now. Extend to ARIMA or Prophet with 6+ months of history.

Exploratory — needs data volume

+ CI/CD Deployment

+ Row-Level Lineage

+ Production Monitoring